

What the codecs keep, and the one thing they lose

Retention of each hand-crafted cue group, as a percentage of what WavLM supports. Mean over 25 partitions.
Teal is better than WavLM, rust is worse. No cell exceeds three points of standard deviation.

Whisper (control)	101%	102%	116%	100%	109%	a second continuous encoder, for reference
Mimi, before	152%	146%	162%	106%	170%	
Mimi, after	137%	143%	143%	96%	161%	
Mimi, histogram	97%	107%	111%	78%	121%	
EnCodec, before	127%	145%	145%	71%	164%	
EnCodec, after	127%	144%	137%	69%	162%	
DAC, before	121%	130%	127%	67%	160%	
DAC, after	115%	127%	117%	61%	156%	
Sylber	80%	90%	80%	81%	87%	variable rate, no fixed clock
DyCAST, before	89%	90%	125%	44%	100%	
DyCAST, after	90%	92%	126%	45%	101%	
	contour	level	voice quality	temporal	spectral	

Figure 4.2 Whisper sits level with WavLM on every group, so the codecs' surplus is a codec property rather than a WavLM deficiency. Temporal is the one group they lose, and DyCAST, which abandons the clock entirely, loses the most of it.